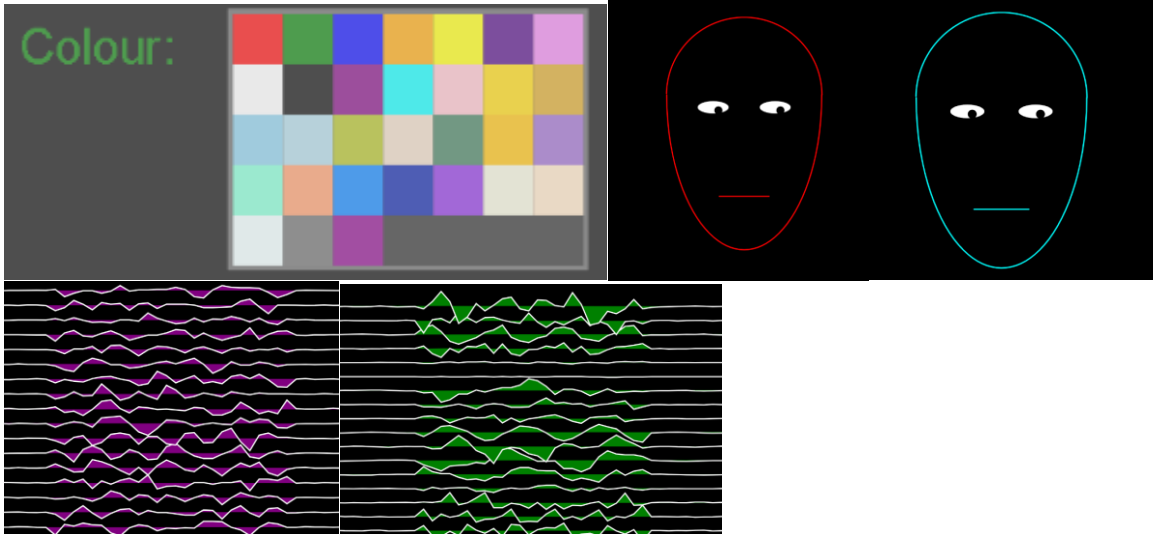


Introduction to Programming II

Project Log

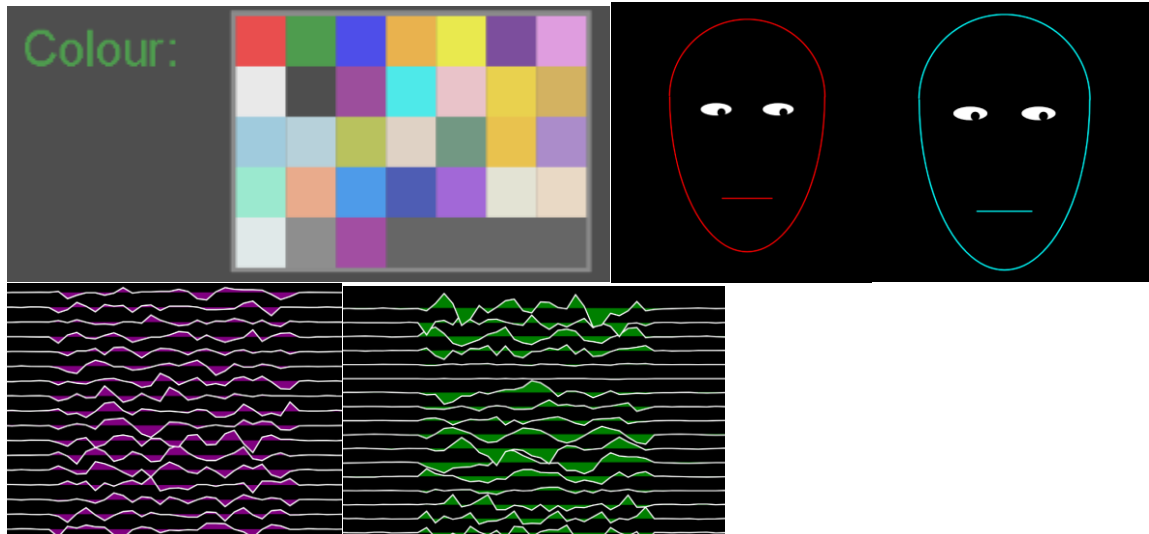
Date: 18/07/2024
Week no: 15 & 16

Project title:	Music Visualizer
Topic:	Color Picker
What progress have you made this topic?	
I have made a color picker in a file named 'colorPicker.js'. This file contains a constructor function which takes 2 variables as inputs and creates a color picker grid using a list of colors. The color picker can be called by visualizers and their location is determined by the input variables. Users can select a color by clicking on any color on the color picker grid and the corresponding element in the visualizer will change based on the input. 	
What problems have you faced and were you able to solve them?	
A problem I faced initially when creating this color picker was drawing each color in a box instead of a line. The first iteration of the code only draws the colors in a single line. 	

What progress have you made this topic?

I have made a color picker in a file named 'colorPicker.js'. This file contains a constructor function which takes 2 variables as inputs and creates a color picker grid using a list of colors.

The color picker can be called by visualizers and their location is determined by the input variables. Users can select a color by clicking on any color on the color picker grid and the corresponding element in the visualizer will change based on the input.



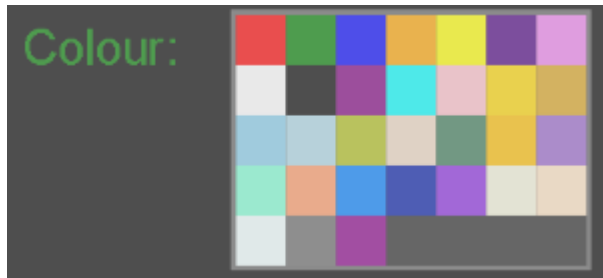
What problems have you faced and were you able to solve them?

A problem I faced initially when creating this color picker was drawing each color in a box instead of a line.

The first iteration of the code only draws the colors in a single line.



I had trouble coming up with a solution to draw the color box in a new line after drawing a certain number color is drawn out. I spent an hour trying different things to make the code draw out the grid and eventually, I managed to make the code draw the color boxes in a new line.



The next problem is its function of selecting a color. My first iteration of the code is a nested for loop, checking for mouseX and mouseY against the x and y value of each color. But this implementation only allows the user to select colors on the first row of the grid.

Code snippet of the initial approach:

```
if(mouseIsPressed){
  var d= -1;
  for(var i=0; i<this.colorList.length; i++){
    if(i%this.gridMul==0){
      d+=1;
    }
    if(mouseX>this.colorPickerX + sizeInterval*i &
      mouseX<this.colorPickerX + sizeInterval + sizeInterval*i &
      mouseY>this.colorPickerY + sizeInterval*d &
      mouseY<this.colorPickerY + sizeInterval + sizeInterval*d
    ){
      this.currentColor = this.colorList[i];
      break;
    }
  }
}
```

Debugging this was a very difficult task as the code often causes the browser to become unresponsive. Therefore, I try to solve this problem with a different approach: calculating the index of the color based on the mouse position.

Code snippet of the new approach:

```
if(mouseIsPressed){
  //calculate which column the mouse is on
  var x = mouseX-(this.colorPickerX);
  x = floor(x/sizeInterval);
  //calculate which row the mouse is on
  var y = mouseY-(this.colorPickerY);
  y = floor(y/sizeInterval);
  //column no. + (row no. * (no. of color box is in a row))
  x += y*this.gridMul;
  //check if index x exists
  if(x<this.colorList.length){
```

```
        this.currentColor = this.colorList[x];  
    }  
}
```

By changing to this approach, I was able to get the color picker to select color as intended, and more efficiently, eliminating the need for many nested for loops within the function.

What are you planning to do over the next few weeks?

Over the next few weeks, I am planning to develop a song queueing system, create a song progress bar for users to skip to a different part of the song as well as the next, previous buttons.



Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?

I am currently on target to successfully complete my project as I managed to complete this implementation successfully without taking time away from implementing other enhancements and functionalities.

Introduction to Programming II

Project Log

Date:1/08/2024

Week no: 17 & 18

Project title:	Music Visualizer
Topic:	Song Queueing system
What progress have you made this topic?	
I have created a song queueing system in a file named 'queueHandler.js'. This file contains a constructor function. This contains functions that queue in a song when an audio file is selected and a few other functions which play the next or previous song, and functions which jump forward or backward a few seconds. I also added an input button in the 'controlsAndInput.js' file to add music files. Choose Files stomper_reggae_bit.mp3 Choose Files 美波 「main actor」 MV.mp3	
What problems have you faced and were you able to solve them?	
When developing the queue handler, I used createFileInput() to create a file selector button to add in a file. Code snippet: <pre>this.addMusic = createFileInput(this.checkFile, "true"); this.checkFile = function(file){ if(file.type === 'audio'){ queueHandlerQ.queue(file); //add names of songs into an array fileNames[sound.length-1] = file.name; } }</pre> This, however, results in an error. I am still unclear on why this causes an error, so I had trouble coming up with a solution. The problem persisted no matter what changes in the code I made. After many trials and errors, I managed to get the application working by creating checkFile as a separate function. Code snippet: <pre>function checkFile(file){</pre>	

```
if(file.type === 'audio'){
  queueHandlerQ.queue(file);
  //add names of songs into an array
  fileNames[sound.length-1] = file.name;
}
}
```

While implementing a function to automatically play the next track in the queue after the current one ends, I encountered another problem. This function did not cause any errors, but did not work as intended.

Code snippet:

```
this.PlayNextSong = function(){
  if(sound[p].currentTime()==sound[p].duration()){
    this.next();
  }
}
```

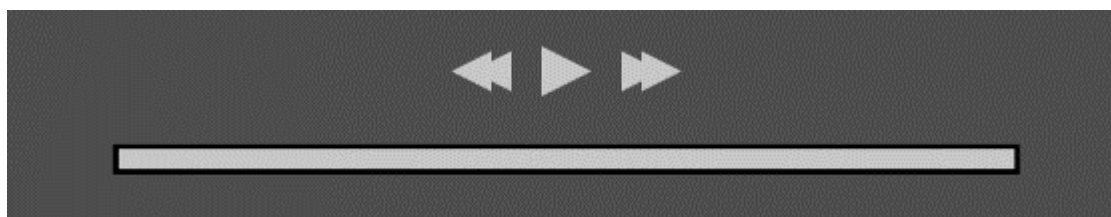
I later discovered that the function ‘.currentTime()’ can return a float with several decimal places. This means I need to make some changes to the code using ‘.round()’ for it to compare integers instead of floats.

Code snippet:

```
this.PlayNextSong = function(){
  if(round(sound[p].currentTime())==round(sound[p].duration())){
    this.next();
  }
}
```

What are you planning to do over the next few weeks?

Over the next few weeks, I am planning to add a progress bar for users to skip to a different part of the song as well as the next, previous buttons to skip to other tracks in the queue.



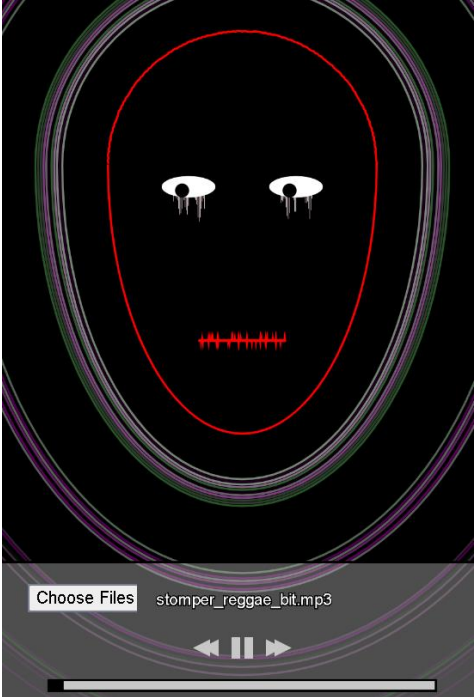
Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?

I am currently on target to successfully complete my project. I was able to finish this implementation and has some spare time, so I started to work on the next implementation.

Introduction to Programming II

Project Log

Date: 15/08/2024
Week no: 19 & 20

Project title:	Music Visualizer
Topic:	Progress Bar, Next/Previous Track, Enhancement for 'face.js'
What progress have you made this topic?	
I added a progress bar next and previous track button, and some enhancement to the extension 'face.js'. I added the waves that spread out around the face when a beat is detected. The progress bar allows users to skip to specific point in the track while the next and previous buttons allow users to play to other tracks in the queue. 	
What problems have you faced and were you able to solve them?	
While implementing the progress bar, I discovered that when paused, the <code>.currentTime()</code> function always returns 0. This means that when pausing a playing track, the progress in the bar becomes empty. <pre data-bbox="1077 1870 1428 1993">87 controlsAndInput.js:67 > sound[p].currentTime() < 0 ></pre>	



To solve this problem, I introduce a global variable that is assigned a value whenever the play/pause is clicked and the track stops playing.



Code snippet:

```
fill(200);
rect(this.progressBarX - w/2,
    this.progressBarY,
    w,
    h
);
//draw the progress of the playing track
fill(0);
if(playing){
    rect(this.progressBarX - w/2,
        this.progressBarY,
        this.CurrentTime,
        h
    );
}
else{
    rect(this.progressBarX - w/2,
        this.progressBarY,
        currTime,
        h
    );
}
```

I also have to add a function that updates this global variable when the song is paused to the ‘.mousePressed’ function as well as the ‘.keyPressed’ function in ‘controlsAndInput.js’ file. This is necessary as these functions checks for input that pause and play the track.

What are you planning to do over the next few weeks?

I am planning to add comments to my code and tidy it up to make it more readable as well as finishing my report with the remaining time.

Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?

I am currently on target to successfully complete my project. I am now left with writing my report submitting my work.